Topology Batch Assignment

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Topology Assignment

Solutions to Questions 5 and 6

Question 5

Let $p : X \to Y$ be an open map. Show that if A is open in X, then the map $q : A \to p(A)$ obtained by restricting p is an open map.

Proof

We are given that $p : X \to Y$ is an open map and that A is an open subset of X.

Consider the restriction of p to A, defined by

$$q : A \longrightarrow p(A), \qquad q(x) = p(x), \quad x \in A.$$

We need to show that q is open.

Note that, a map is open if the image of every open subset of its domain is open in its codomain.

Let U be an arbitrary open subset of A.

Since U is open in the subspace topology on A, there exists an open subset V of X such that

$$U = A \cap V.$$

Now, both A and V are open in X. Therefore, their intersection is also open in X. Hence, U is open in X.

Since $p : X \to Y$ is an open map, the image of every open subset of X is open in Y. Consequently, $p(U)$ is open in Y.

On the other hand, since $U \subseteq A$, we have

$$p(U) \subseteq p(A).$$

Also, by the definition of the restriction q, we obtain

$$q(U) = p(U).$$

Thus, $q(U)$ is an open subset of Y contained in $p(A)$.

By the definition of the subspace topology, $q(U)$ is therefore open in $p(A)$.

Since U was an arbitrary open subset of A, the image of every open subset of A under q is open in $p(A)$.

Therefore,

$$\boxed{q : A \to p(A) \text{ is an open map.}}$$

Question 6

Let $\mathbb{R}_K$ denote the real line with the K-topology, where

$$K = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Let Y be the quotient space obtained from $\mathbb{R}_K$ by identifying all points of K to a single point, and let

$$p : \mathbb{R}_K \longrightarrow Y$$

be the quotient map.

Recall that a basis for the K-topology consists of sets of the form

$$(a, b) \qquad \text{and} \qquad (a, b) \setminus K,$$

where $a, b \in \mathbb{R}$ and $a < b$.

(a) Show that Y satisfies the T_1 axiom, but is not Hausdorff.

Proof

We shall prove the two assertions separately.

Step 1: Prove that Y satisfies the T_1 axiom.

Recall that a topological space is said to satisfy the T_1 axiom if every singleton subset is closed.

Under the quotient map p, all the points of K are identified to a single point. Let this point be denoted by $y_0 = p(K)$.

Thus, the equivalence classes are

$$p^{-1}(\{y_0\}) = K$$

and

$$p^{-1}(\{p(x)\}) = \{x\}, \qquad x \in \mathbb{R} \setminus K.$$

We first observe that K is closed in $\mathbb{R}_K$.

Indeed, $\mathbb{R} \setminus K$ is open in the K-topology. To see this, note that every point outside K has a basic neighbourhood contained in $\mathbb{R} \setminus K$.

For the point 0, we can take a neighbourhood of the form

$$(-\varepsilon, \varepsilon) \setminus K.$$

For any other point outside K, we can choose a sufficiently small ordinary open interval avoiding K.

Hence, K is closed in $\mathbb{R}_K$.

Also, every singleton $\{x\}$ is closed in $\mathbb{R}_K$, since the K-topology is finer than the usual topology on $\mathbb{R}$, which is a T_1 space.

Now, consider an arbitrary point $y \in Y$.

There are two possibilities:

Case 1: If $y = y_0 = p(K)$, then

$$p^{-1}(\{y\}) = K,$$

which is closed in $\mathbb{R}_K$.

Case 2: If $y = p(x)$ for some $x \in \mathbb{R} \setminus K$, then

$$p^{-1}(\{y\}) = \{x\},$$

which is also closed in $\mathbb{R}_K$.

Therefore, in both cases, $p^{-1}(\{y\})$ is closed in $\mathbb{R}_K$.

By the definition of the quotient topology, a subset F of Y is closed if and only if $p^{-1}(F)$ is closed in $\mathbb{R}_K$.

Consequently, $\{y\}$ is closed in Y.

Since y was arbitrary, every singleton subset of Y is closed.

Thus,

$$\boxed{Y \text{ satisfies the } T_1 \text{ axiom.}}$$

Step 2: Prove that Y is not Hausdorff.

Recall that a topological space is Hausdorff if every pair of distinct points can be separated by disjoint open neighbourhoods.

We shall show that the two distinct points

$$y_0 = p(K) \qquad \text{and} \qquad y_1 = p(0)$$

cannot be separated by disjoint open neighbourhoods.

Notice that $0 \notin K$, so $y_0 \neq y_1$.

Suppose, on the contrary, that Y is Hausdorff.

Then there exist disjoint open neighbourhoods U and V of y_0 and y_1, respectively. Thus,

$$y_0 \in U, \qquad y_1 \in V, \qquad U \cap V = \varnothing.$$

Since p is a quotient map, the sets $p^{-1}(U)$ and $p^{-1}(V)$ are open in $\mathbb{R}_K$.

Furthermore, since $y_0 \in U$ and $p^{-1}(\{y_0\}) = K$, we have

$$K \subseteq p^{-1}(U).$$

Similarly, since $y_1 = p(0) \in V$, we have

$$0 \in p^{-1}(V).$$

Now, $p^{-1}(V)$ is an open neighbourhood of 0 in the K-topology.

By the definition of the K-topology, every open neighbourhood of 0 contains a set of the form

$$(-\varepsilon, \varepsilon) \setminus K$$

for some $\varepsilon > 0$.

Therefore,

$$(-\varepsilon, \varepsilon) \setminus K \subseteq p^{-1}(V). \tag{1}$$

Since

$$\frac{1}{n} \longrightarrow 0 \qquad \text{as } n \longrightarrow \infty,$$

we can choose n sufficiently large so that

$$0 < \frac{1}{n} < \frac{\varepsilon}{2}.$$

Put $k = \frac{1}{n}$. Then $k \in K$ and

$$k \in p^{-1}(U).$$

Since $p^{-1}(U)$ is open in $\mathbb{R}_K$ and contains k, there exists $\delta > 0$ such that

$$(k - \delta, k + \delta) \setminus K \subseteq p^{-1}(U). \tag{2}$$

By choosing δ sufficiently small, we may also assume that

$$\delta < \varepsilon - k.$$

Consequently,

$$(k - \delta, k + \delta) \subseteq (-\varepsilon, \varepsilon).$$

Since every nonempty open interval contains points outside K, we can choose

$$z \in (k - \delta, k + \delta) \setminus K.$$

Using (2), we obtain

$$z \in p^{-1}(U).$$

Also, since $z \in (-\varepsilon, \varepsilon) \setminus K$, relation (1) gives

$$z \in p^{-1}(V).$$

Therefore,

$$p^{-1}(U) \cap p^{-1}(V) \neq \varnothing.$$

But this is impossible, because inverse images preserve intersections and

$$\begin{aligned} p^{-1}(U) \cap p^{-1}(V) &= p^{-1}(U \cap V) \\ &= p^{-1}(\varnothing) \\ &= \varnothing. \end{aligned}$$

We have obtained a contradiction.

Hence, the two distinct points $p(K)$ and $p(0)$ cannot be separated by disjoint open neighbourhoods.

Therefore,

$$\boxed{Y \text{ is not Hausdorff.}}$$

Combining the two steps, we conclude that

$$\boxed{Y \text{ is } T_1 \text{ but is not Hausdorff.}}$$

(b) Show that $p \times p : \mathbb{R}_K \times \mathbb{R}_K \to Y \times Y$ is not a quotient map.

Proof

We shall use the fact that the diagonal of a topological space is closed in its product with itself if and only if the space is Hausdorff.

Since, by part (a), Y is not Hausdorff, its diagonal

$$\Delta_Y = \{(y, y) : y \in Y\}$$

is not closed in $Y \times Y$.

Now, consider the inverse image of Δ_Y under the map $p \times p$.

By definition,

$$\begin{aligned}
&(p \times p)^{-1}(\Delta_Y) \\
&= \{(x_1, x_2) \in \mathbb{R}_K \times \mathbb{R}_K : (p(x_1), p(x_2)) \in \Delta_Y\} \\
&= \{(x_1, x_2) \in \mathbb{R}_K \times \mathbb{R}_K : p(x_1) = p(x_2)\}.
\end{aligned}$$

Recall that two points of $\mathbb{R}_K$ have the same image under p precisely when either they are equal or both belong to K.

Therefore,

$$\boxed{(p \times p)^{-1}(\Delta_Y) = \Delta_{\mathbb{R}_K} \cup (K \times K),}$$

where

$$\Delta_{\mathbb{R}_K} = \{(x, x) : x \in \mathbb{R}\}$$

is the diagonal of $\mathbb{R}_K \times \mathbb{R}_K$.

We now show that this inverse image is closed in $\mathbb{R}_K \times \mathbb{R}_K$.

First, observe that $\mathbb{R}_K$ is Hausdorff because its topology is finer than the usual topology on $\mathbb{R}$, which is Hausdorff.

It is a standard result that the diagonal of a Hausdorff space is closed in its product with itself.

Hence,

$$\Delta_{\mathbb{R}_K}$$

is closed in $\mathbb{R}_K \times \mathbb{R}_K$.

Secondly, from part (a), we know that K is closed in $\mathbb{R}_K$.

Therefore, by the product topology,

$$K \times K$$

is closed in $\mathbb{R}_K \times \mathbb{R}_K$.

Since the union of finitely many closed sets is closed, we conclude that

$$\Delta_{\mathbb{R}_K} \cup (K \times K)$$

is closed in $\mathbb{R}_K \times \mathbb{R}_K$.

Thus,

$$\boxed{(p \times p)^{-1}(\Delta_Y) \text{ is closed in } \mathbb{R}_K \times \mathbb{R}_K.}$$

Finally, recall the defining property of a quotient map.

A surjective map $f : X \to Z$ is a quotient map if and only if, for every subset $B \subseteq Z$,

$$B \text{ is closed in } Z \iff f^{-1}(B) \text{ is closed in } X.$$

In particular, if $p \times p$ were a quotient map, the closedness of $(p \times p)^{-1}(\Delta_Y)$ would imply that Δ_Y is closed in $Y \times Y$.

However, we have already established that Δ_Y is not closed, since Y is not Hausdorff.

This is a contradiction.

Therefore,

$$\boxed{\begin{array}{c} p \times p : \mathbb{R}_K \times \mathbb{R}_K \longrightarrow Y \times Y \\ \text{is not a quotient map.} \end{array}}$$